

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/251881129>

Differential rumen microbial dynamics and fermentation parameters in cattle fed high fibre and high concentrate diet.

Article · January 2013

CITATIONS

8

READS

222

6 authors, including:



Sunil Sirohi

National Dairy Research Institute

118 PUBLICATIONS 1,231 CITATIONS

[SEE PROFILE](#)



Sumit Singh Dagar

Agharkar Research Institute

90 PUBLICATIONS 1,885 CITATIONS

[SEE PROFILE](#)



Prem Prashant Chaudhary

National Institutes of Health

60 PUBLICATIONS 923 CITATIONS

[SEE PROFILE](#)



Anil Kumar Puniya

ICAR-National Dairy Research Institute, Karnal - 132001 (Haryana), India

249 PUBLICATIONS 5,883 CITATIONS

[SEE PROFILE](#)

Differential Rumen Microbial Dynamics and Fermentation Parameters in Cattle Fed on High Fibre and High Concentrate Diets

Sunil Kumar Sirohi^{*1}, Sumit Singh Dagar¹, Nasib Singh¹, Prem Prashant Chaudhary¹,
Anil Kumar Puniya² and Dheer Singh³

¹Nutrition Biotechnology Lab, Dairy Cattle Nutrition Division, ²Dairy Microbiology Division

³Animal Biochemistry Division, National Dairy Research Institute, Karnal-132001, Haryana, India

ABSTRACT

In the present study, the effect of two diets containing varying level of fibre was evaluated on the rumen microbial dynamics and rumen fermentation characteristics at different intervals in Karan Fries crossbred cattle. Ruminal microbial groups were quantified using species specific primers using real-time PCR assays. A significant increase in copy numbers of fibrolytic bacteria viz. *Ruminococcus albus*, *R. flavefaciens*, *Butyrivibrio fibrisolvens* and *Fibrobacter succinogens* was observed on day 11 in rumen of cattle fed on high fibre diet (HFD). Similarly, the total bacterial and total anaerobic fungi population also increased by nearly 50% on day 11 in HFD group than day 0. However, the levels of acetate, propionate and butyrate as well as ammonia nitrogen (NH₃-N) were found significantly higher in cattle fed high concentrate diet (HCD) compared to HFD group. These findings indicate the importance of fibre content of the diet influencing microbial dynamics in rumen ecosystem.

Keywords : High concentrate diet, High fibre diet, Rumen bacteria, Rumen fungi, Real-time PCR, Cattle

INTRODUCTION

A highly complex and diverse microbiota of the rumen comprising bacteria, archaea, fungi and protozoa is involved in the breakdown of feed components to fulfill most of the host's energy needs. The adaptability of the ruminal microbial community allows ruminant animals to consume a wide variety of feedstuffs. The type and amount of feed materials consumed by the host affect the nutritional supply to ruminal microbes and the end products synthesized, thereby influencing the nutrients absorbed by the host. Forage is an important component of diets fed to the ruminants. Fibrolytic rumen bacteria play a major role in the breakdown of cellulose, hemicellulose, and other fibrous components of the forage, thus allowing the animals to digest the diet and to harness the energy present within. Feeding forages that are easier for fibrolytic bacteria to degrade and that encourage colonization by certain bacterial species could potentially increase the amount of fiber utilization.

Numerous factors can alter ruminal microbial population such as dietary fat supplements (Varadyova et al., 2008), forage to concentrate ratio, forage source (Vlaeminck et

al., 2006), and dietary additives. Previous research has demonstrated that the microbial diversity and activities can be influenced by modifications to the diet (Hernandez-Sanabria et al., 2012). However, our understanding of how rumen microbial numbers fluctuate and adapt to different diets is still in its infancy. Estimating population sizes in the rumen is difficult because microbial population fluctuate considerably between animals, pre or post feeding, and because of sample heterogeneity. The composition of the diet fed to ruminants has been reported to change the composition of the bacterial communities in the fluid and food particle fractions (Tajima et al., 2001; Koike et al., 2003).

Real-time PCR (RT-PCR) is an approach that allows continuous monitoring of PCR product formation. It has been used to monitor bacterial, protozoal, methanogen populations in the rumen of different domesticated and wild ruminants (Skillman et al., 2006; Stevenson and Weimer, 2007; Masoni et al., 2007). RT-PCR was used in several studies to quantify the abundance of particular bacterial species in the rumen under a variety of conditions (Tajima et al., 2001; Klieve

^{*}Corresponding author E-mail : sirohisk@gmail.com

et al., 2003). In this study, we hypothesized that fibrolytic bacterial species and total bacterial or fungal populations fluctuate depending on the content of fiber in the diet. We utilized RT-PCR to quantify the total bacteria, total anaerobic fungi, and four major groups of fibrolytic bacteria i.e. *Fibrobacter succinogenes*, *Ruminococcus flavefaciens*, *Ruminococcus albus* and *Butyrivibrio fibrisolvens* in cattle during shift from high fibre diet to high concentrate diet.

MATERIALS AND METHODS

Four fistulated male cattle (Karan Fries; mean body weight 455 ± 5 kg) were selected for study and feeding trial was conducted in 2×2 switch over design. Animal were fed on high fibre diet (HFD) containing 80% roughage (40% wheat straw, 40% maize fodder) and 20% concentrate and high concentrate diet (HCD) containing 80% concentrate and 20% roughage (10% wheat straw; 10% maize fodder). Feeding trials were conducted for period of 30 days and then switch over to another diet after feeding normal diet (60% roughage and 40 % concentrate) for 7 days period to remove the carryover effect. During the trial, animals were

individually accommodated in animal sheds located at National Dairy Research Institute, Karnal, India.

The rumen samples were collected on day 0, 11 and 21 days after initiation of feeding trial. Required representative samples of rumen contents were drawn prior to the morning feeding through a plastic stomach tube into a warm thermos flask previously flushed with O_2 -free CO_2 and samples were immediately brought to the laboratory for further analysis.

The samples were homogenized using blender and squeezed through four layers of muslin cloth to remove feed particles and processed for DNA extraction as described previously (Chaudhary and Sirohi, 2009; Chaudhary et al., 2011). Total genomic DNA was extracted and the concentrations were measured at 260 nm with a Nanodrop. In order to minimize animal to animal variations, extracted DNA from all the animals was pooled by mixing in equal volume before further processing. Then PCR amplification was performed in MyCyclerTM thermal cycler (Bio-Rad, USA) as described by Chaudhary and Sirohi (2009). The PCR conditions were 95°C for 5 min, followed by 35 cycles

Table 1. Chemical composition and dry matter intake of feeds offered to Karan Fries cattle

Particular	Concentrate	Green maize	Wheat straw
	Composition of the diet		
Dry matter	90.00	14.94	90.30
Organic matter	90.11	90.75	89.73
Crude protein	20.80	9.77	3.13
Ether extract	4.10	1.55	1.01
Total Ash	9.89	9.25	10.21
Neutral Detergent Fibre	43.41	65.36	77.42
Acid Detergent Fibre	15.28	35.36	47.53
Hemi-cellulose	28.13	30.0	29.89
Particular	Dry matter intake (kg/d/animal)		
	HFD	HCD	
Average body weight (kg)	450	455	
Wheat straw	2.84	0.7	
Concentrate	1.42	5.7	
Maize	2.84	0.7	
Total DMI	7.1	7.1	

consisting of 95°C for 30 sec, various annealing temperatures for 30 sec, 72°C for 1 min, and a final extension period of 72°C for 10 min.

To establish a quantitative assay, all the targeted groups were amplified using specific primers (Table 3) and cloned by using Stratagene Blunt End cloning kit (Stratagene, USA) and purified by QIAprep kit (Qiagen, Venlo, The Netherlands). The purified plasmids were quantified by Nano-drop spectrophotometer (Thermo Fischer, USA) with multiple dilutions. The target DNA was quantified by using 10-fold serial dilutions from 10⁻¹ to 10⁻⁸ plasmid copies of the previously quantified plasmid standards. Real time PCR amplifications and detections were performed in a MJ Mini™ Mini Opticon Real-Time PCR System (Bio-Rad, USA) as described previously (Chaudhary et al., 2011). In brief, SYBR Green master mix (Fermentas, USA) was used for PCR amplification. Samples were assayed 25 µl reaction mixture containing 5 mM MgCl₂, 12.5 µl of 10X master mix (including Fast start enzyme, Fast start Taq DNA polymerase, reaction buffer, dNTP mixture, MgCl₂, and SYBR green dye), 50 ng of template DNA, and 0.5 µM of each primer. All PCR reactions were performed in triplicate.

The chemical composition (organic matter, crude protein, ether extract, total ash) of substrate was carried out as per the methods of AOAC (2005). The neutral

detergent fibre, acid detergent fiber and hemicelluloses of substrates were determined according to method described earlier (Van Soest et al., 1991).

Rumen liquor was centrifuged at 3000 rpm for 10 min and IVFA were estimated by taking 1 ml of the supernatant treated with 25% metaphosphoric (4 ml) prepared in 1N H₂SO₄ and kept for 3-4 h at ambient temperature (Erwin et al., 1961). Samples were centrifuged at 3000 rpm for 10 min and clear supernatants were stored at -20°C until analyzed. IVFA was estimated using gas chromatograph (Nucon 5700, India) equipped with flame ionization detector (FID) and stainless steel column (length 4'; o.d ¼"; i.d 3 mm) packed with Chromosorb- 101. Temperature of injection port, column and detector was set at 200, 180 and 210°C, respectively. The flow rate of carrier gas (nitrogen) through the column was 40 ml/ min and the flow rates of hydrogen gas and air through FID were 30 and 300 ml/ min, respectively. Sample (2 µl) was injected through the injection port using 10 µl capacity Hamilton syringe. IVFAs of the samples were identified on the basis of their retention time and their concentrations were calculated by comparing the retention time as well as the peak area of standards.

Rumen liquor (5 ml) was mixed with 1 N NaOH (2 ml) and the steam was passed over it in KELPLUS - Nitrogen analyzer (Pelican, India). NH₃ evolved during this process was collected in boric acid solution contain-

Table 2. Rumen fermentation parameters of cattle fed with high fibre (HFD) and high concentrate diets (HCD)

Days	Acetate (mM/ 100 ml)		Propionate (mM/ 100 ml)		Butyrate (mM/ 100 ml)		NH ₃ -N (mg/100 ml)	
	HFD	HCD	HFD	HCD	HFD	HCD	HFD	HCD
Initial (0 d)	11.67±0.51	11.87±0.61	2.11±0.02	2.21±0.02	1.14±0.18	1.22±0.16	9.29±0.36	9.34±0.32
11	13.88±0.23	14.13±0.89	2.70±0.05	3.35±0.59	1.62±0.03	2.29±0.01	11.48±1.61	13.86±0.24
21	14.30±0.21	13.48±1.62	3.02±0.04	3.12±0.36	1.63±0.03	1.86±0.18	11.24±0.13	17.22±1.69
Average	13.28	13.09	2.61	2.86	1.46	1.76	10.65	13.44
Least Significant difference								
Diet (D)	1.60		0.66		0.32		2.86	
Periods (P)	NS		NS		0.51		3.17	
D × P	NS		NS		*		*	

* Significant at P<0.05, NS : Not significant

ing mixed indicator and titrated against N/100 H₂SO₄.

Experimental data were analyzed in 2 × 2 switch over design for analysis of variance as per Snedecor and Cochran (1968). When the overall *F*-test was significant, differences between means and the control were declared significant at *P* < 0.05 using the Fisher's least significant difference (Critical difference).

RESULTS AND DISCUSSION

The chemical composition of feeds offered and rumen fermentation parameters were presented in Table 1 and 2. Concentrations of VFAs and NH₃-N were determined in rumen fluid samples of cattle to assess the interrelationship between important rumen microbial groups and its functions in the rumen of cattle fed with HFD and HCD formulated using green fodder, concentrate and wheat straw. Our findings indicated that IVFA and NH₃-N varied significantly between two experimental groups fed HFD and HCD diets. The dietary change affected the rumen environment of Karan Fries cattle as evident by increase in the NH₃-N concentration from 9.34±0.32 on day 0 to 13.86±0.24 on day 11 in experimental animals fed on HCD (Table 2).

By day 21, this increase was to the tune of about 95%. In comparison, levels of NH₃-N in animals fed on HFD did not show appreciable increase as in HCD. In terms of IVFAs, cattle fed on HCD exhibited higher increase in propionate and butyrate compared to those fed on HFD. Interestingly, no such variation was found between two experimental groups with regard to acetate concentration. The increase in propionate in HCD fed animals due to increase in the availability of highly digestible nutrients as compared to HFD. The variability in chemical composition of diets clearly affects the fermentation pattern in rumen.

Change in population of different groups fibrolytic bacteria, total bacteria and total fungi in cattle on day 0, 11 and 21 of feeding trial were quantified by RT-PCR. The copy numbers of *R. albus* exhibited a steady increase from day 0 (1.35 × 10⁵) to day 21 (1.69 × 10⁵) / 50 ng of DNA in cattle fed on HFD (Table 4). The increase in copy number of *F. succinogenes* was most pronounced as evident by 100% increase on day 11 in HFD group. However, its number declined to 1.35 × 10⁵/50ng of DNA by day 21. The copy numbers of *B. fibrisolvens* and *R.*

Table 3. Primer used for real-time PCR quantification of rumen microbial population in Karan Fries cattle

Bacterial species/ group	Primer sequence	Product size (bp)	Annealing Temp (°C)	Refer- ences
Total bacteria	Fwd- 5'-CGGCAACGAGCGCAACCC-3' Rev- 5'-CCATTGTAGCACGTGTGTAGCC-3'	145	60.2	Denman and McSweeney (2006)
Total anaerobic fungi	Fwd- 5'-GAGGAAGTAAAAGTCGTAACAAGGTTTC-3' Rev- 5'-CAAATTCACAAAGGGTAGGATGATT-3'	110	58.4	Denman and McSweeney (2006)
<i>Ruminococcus albus</i>	Fwd- 5'-CCCTAAACAGTCTTAGTTCG-3' Rev- 5'-CCTCCTTGCGGTTAGAACA-3'	175	56.0	Tajima et al., (2001)
<i>Ruminococcus flavefaciens</i>	Fwd- 5'-CGAACGGAGATAATTTGAGTTTACTTAGG-3' Rev- 5'-CGGTCTCTGTATGTTATGAGGTATTA-3'	132	56.3	Denman and McSweeney (2006)
<i>Butyrivibrio fibrisolvens</i>	Fwd- 5'-TCTGGAAACGGATGGTA-3' Rev- 5'-CCTTTAAGACAGGAGTTTACAA-3'	284	50.0	Forster et al., 1996)
<i>Fibrobacter succinogenes</i>	Fwd- 5'-GTTCGGAATTACTGGGCGTAAA-3' Rev- 5'-CGCCTGCCCTGAACTATC-3'	121	60.6	Denman and McSweeney (2006)

Table 4. Real-time PCR measurement of relative abundance of individual fibrolytic bacteria, total bacteria and anaerobic fungi in rumen of Karan Fries cattle fed with high fibre (HFD) and high concentrate diet (HCD)

Microbes species or group (Copy number per 50 ng of DNA)	High fibre diet (80% roughage : 20% concentrate)				High concentrate diet (20% roughage : 80% concentrate)				Significance (Least significant difference)	
	Day 0	Day 11	Day 21	Average	Day 0	Day 11	Day 21	Average	Diets	Periods
	Day 0	Day 11	Day 21	Average	Day 0	Day 11	Day 21	Average	Diets	D×P
<i>Ruminococcus albus</i> (×10 ⁵)	1.35	1.58	1.69	1.54	1.40	1.29	1.27	1.32	0.20	NS
<i>Ruminococcus flavefaciens</i> (×10 ⁷)	1.03	1.28	1.01	1.11	1.13	0.95	0.97	1.01	0.12	**
<i>Butyrivibrio fibrisolvens</i> (×10 ⁷)	0.97	1.38	1.06	1.14	0.99	0.87	0.89	0.91	0.20	*
<i>Fibrobacter succinogenes</i> (×10 ⁵)	1.11	2.26	1.35	1.57	1.18	1.03	1.04	1.08	0.50	*
Total bacteria (×10 ⁸)	4.93	7.46	4.16	5.52	5.00	3.52	3.97	4.16	1.40	***
Total fungi (×10 ⁵)	6.87	8.22	6.64	7.25	6.95	6.37	6.30	6.54	0.70	**

* Significant at P<0.05 ; **Significant at P<0.01 ; ***Significant at P<0.001, NS : Not Significant

flavefaciens exhibited an increase of approximately 40 and 25%, respectively on day 11. However, their numbers slides to initial (day 0) levels on day 21. Similar trend was observed in case of total bacteria and total rumen fungi. Compared to HFD group, the copy numbers of various bacterial species as well total bacteria and fungi did not changed significantly from the day 0 levels during the course of feeding trials (Table 4). Overall, it was observed that feeding of cattle with fibrous diet stimulate the populations of those microbial species and groups that are involved in fibre breakdown.

It has been demonstrated that the composition of the diet fed to ruminants modulate the microbial communities present in the rumen (Tajima et al., 2001; Koike et al., 2003; Larue et al., 2005). However, the adaptation of the bacterial communities during diet changes remains less understood. How the number of different microbial groups changed in rumen on different diets also needs to be studied for developing the feeding strategies for different category of animals. In recent times, molecular methods have gained popularity for estimating microbial population in rumen environment opened new dimensions for studying these microbial entities.

Our findings of the present study indicated a increase in VFA, particularly, propionate, butyrate and NH₃-N levels in animals fed on HCD compared to those given HFD. The increase in VFAs could be attributed to the increase of starch or rapidly digestible carbohydrates as also reported in the experiment with corn-silage-based diet, shown by Calsamiglia et al., (2008). The large decrease in the NH₃-N content of the rumen after the dietary change could be explained by the increased availability of soluble nitrogen in HCD as compared with HFD.

In this study, the effect of dietary change from standard diet (40: 60; concentrate : roughage) or initial values to 20: 80 (concentrate : roughage) on fibrolytic bacteria, total bacteria and anaerobic fungi were significant. Our findings are in nice correlation with previous studies related to decipher the relationship between dietary composition and ruminal micro biota (Tajima et al., 2000 and 2001; Kocherginskaya et al.,

2001). Tajima et al., (2000) reported that the bacterial communities before and after a switch from a high-fibre diet to a high-grain diet revealed considerable changes in bacterial composition after 3 days and some characteristics of the resilient behavior of the bacterial communities after 28 days. Koike and Kobayashi (2001) and Kongmun et al., (2010) found that *F. succinogenes* was slightly more abundant than *R. albus* and *R. flavefaciens* in sheep and buffalo. Therefore, the variations between cattle and buffalo in the proportions and numbers of ruminal predominant bacteria and fungi might account for the differences in intake and fermentation end products obtained in current study. Further, the effect of several diet regimen and diet shift had reported in different ruminant species. Prakash et al., (2013) determined the effect of different diets on fermentation and microbial population in mithun (*Bos frontalis*). An increase in the population of *Fibrobacter succinogenes*, *Ruminococcus flavefaciens* and total bacteria was observed in mithun fed 50% concentrate mixture compared to those fed only tree foliages. Further studies are warranted to study rumen ecosystem over a more comprehensive time point range and bigger dietary differences to enhance our understanding of rumen function, including relationships between rumen fermentation parameters and the microbial community.

CONCLUSION

It was concluded from the study that high fibre containing diet enhances the number of fibrolytic bacteria and rumen fungi whereas concentrate rich diet favors other ruminal microbial groups and also increases acetate, propionate, butyrate and $\text{NH}_3\text{-N}$. These findings establish the importance of diet composition in the fermentation pattern of ruminants.

ACKNOWLEDGEMENTS

We gratefully acknowledge the financial support received from the National Agricultural Innovation Project (NAIP), ICAR, Government of India.

REFERENCES

- AOAC. 1995. Official Methods of Analysis, 16th ed. Animal Feeds: Association of Official Analytical Chemists, VA, USA.
- Calsamiglia, S., Cardozo, P.W., Ferret, A. and Bach, A. 2008. Changes in rumen microbial fermentation are due to a combined effect of type of diet and pH. *J. Anim. Sci.* 86: 702-711
- Chaudhary, P.P. and Sirohi, S.K. 2009. Dominance of *Methanomicrobium* phylotype in methanogen population present in Murrah buffaloes (*Bubalus bubalis*). *Lett. Appl. Microbiol.* 49: 274-277
- Chaudhary, P.P., Sirohi, S.K. and Saxena, J. 2012. Diversity analysis of methanogens in rumen of *Bubalus bubalis* by 16S ribotyping and sequence analysis. *Gene* 493: 13-17
- Chaudhary, P.P., Sirohi, S.K., Singh, D. and Saxena, J. 2011. Methyl coenzyme M reductase (*mcrA*) gene based phylogenetic analysis of methanogens population in Murrah buffaloes (*Bubalus bubalis*). *J. Microbiol.* 49: 558-561
- Denman, S.E. and McSweeney, C.S. 2006. Development of a real-time PCR assay for monitoring anaerobic fungal and cellulolytic bacterial populations within the rumen. *FEMS Microbiol Ecol* 58: 572–582
- Erwin, E.S., Macro, G.A. and Emery, E.M. 1961. Volatile fatty acid analysis of blood and rumen fluid by Gas chromatography. *J. Dairy Sci.* 44: 1768-1771
- Forster, R.J., Teather, R.M., Gong, J. and Deng, S.J. 1996. 16S rDNA analysis of *Butyrivibrio fibrisolvens* : phylogenetic position and relation to butyrate-producing anaerobic bacteria from the rumen of white-tailed deer. *Lett. Appl. Microbiol.* 23: 218-222
- Hernandez-Sanabria, E., Goonewardene, L.A., Wang, Z., Durunna, O.N., Moore, S.S. and Guan, L.L. 2012. Impact of feed efficiency and diet on adaptive variations in the bacterial community in the rumen fluid of cattle. *Appl. Environ. Microbiol.* 78: 1203-1214
- Klieve A.V., Hennessy, D., Ouwerkerk, D., Forster, R.J., Mackie, R.I. and Attwood, G.T. 2003. Establishing populations of *Megasphaera elsdenii* YE34 and *Butyrivibrio fibrisolvens* YE44 in the rumen of cattle fed high grain diets. *J. Appl. Bacteriol.* 95: 621–630
- Kocherginskaya, S.A., Aminov, R.I. and White, B.A. 2001. Analysis of the rumen bacterial diversity under two different diet conditions using denaturing gradient gel electrophoresis, random sequencing, and statistical ecology approaches. *Anaerobe* 7: 119–134
- Koike, S. and Kobayashi, Y. 2001. Development and use of competitive PCR assays for the rumen cellulolytic bacteria: *Fibrobacter succinogenes*, *Ruminococcus albus* and *Ruminococcus flavefaciens*. *FEMS Microbiol.*

- Lett. 204: 361-366
- Koike, S., Pan, J., Kobayashi, Y. and Tannaka, K. 2003. Kinetic of in sacco fiber attachment of representative ruminal cellulolytic bacteria monitoring by competitive PCR. J. Dairy Sci. 86: 1429-1435
- Kongmun, P., Wanapat, M., Pakdee, P. and Navanukraw, C. 2010. Effect of coconut oil and garlic powder on in vitro fermentation using gas production technique. Livestock Sci. 12: 38-44
- Koike S., Yabuki, H. and Kobayashi, Y. 2007. Validation and application of real-time PCR assays for representative rumen bacteria. J. Anim. Sci. 78: 135-141
- Larue, R., Yu, Z., Parisi, V.A., Egan, A.R. and Morrison, M. 2005. Novel microbial diversity adherent to plant biomass in the herbivore gastrointestinal tract, as revealed by ribosomal intergenic spacer analysis and rrs gene sequencing. Environ. Microbiol. 7: 530-543
- Mosoni, P., Durand, F.C., Maillet, C.B. and Forano, E. 2007. Quantification by real-time PCR of cellulolytic bacteria in the rumen of sheep after supplementation of a forage diet with readily fermentable carbohydrates: effect of a yeast additive. Appl. Microbiol. 103: 2676-2685
- Prakash, B., Saha, S.K., Khate, K., Agarwal, N., Katole, S., Haque, N. and Rajkhowa, C. 2013. Rumen microbial variation and nutrient utilisation in mithun (*Bos frontalis*) under different feeding regimes. J. Anim. Physiol. Anim. Nutr. (Berl). 97: 297-304.
- Sirohi, S.K., Singh, N., Dagar, S.S. and Puniya, A.K. 2012. Molecular tools for deciphering the microbial community structure and diversity in rumen ecosystem. Appl. Microbiol. Biotechnol. 95: 1135-1154
- Skillman, L.C., Toovey, A.F., Williams, A.J. and Wright, A.D. 2006. Development and validation of a real-time PCR method to quantify rumen protozoa and examination of variability between *Entodinium* populations in sheep offered a hay-based diet. Appl. Microbiol. Biotechnol. 72: 200-206
- Stevenson, D.M. and Weimer, P.J. 2007. Dominance of *Prevotella* and low abundance of classical ruminal bacterial species in the bovine rumen as revealed by relative quantification real-time PCR. Appl. Microbiol. Biotechnol. 75: 165-174
- Tajima, K., Aminov, R.I., Nagamine, T., Matsui, H., Nakamura, M. and Benno, Y. 2001. Diet-dependent shifts in the bacterial population of the rumen revealed with real-time PCR. Appl. Environ. Microbiol. 67: 2766-2774
- Tajima, K., Aminov, R.I., Nagamine, T., Ogata, K., Nakamura, M., Matsui, H. and Benno, Y. 1999. Rumen bacterial diversity as determined by sequence analysis of 16S rDNA libraries. FEMS Microbiol. Ecol. 29: 159-169
- Tajima, K., Arai, S., Ogata, K., Nagamine, T., Mastui, H. and Nakamura, M. 2000. Rumen bacterial community transition during adaptation to high-grain diet. Anaerobe 6: 273-284
- Van Soest, P.J., Robertson, J.B. and Lewis, B.A. 1991. Methods for dietary fiber, neutral detergent fiber, and non-starch polysaccharides in relation to animal nutrition. J. Dairy Sci. 74: 3583-3597
- Varadyova, Z., Kisidayova, S., Siroka, P. and Jalc, D. 2008. Comparison of fatty acid composition of bacterial and protozoal fractions in rumen fluid of sheep fed diet supplemented with sunflower, rapeseed and linseed oils. Anim. Feed Sci. Technol. 144: 44-54
- Vlaeminck, B., Fievez, V., Demeyer, D. and Dewhurst, R.J. 2006. Effect of forage: concentrate ratio on fatty acid composition of rumen bacteria isolated from ruminal and duodenal digesta. J. Dairy Sci. 89: 2668-2678
- Wanapat, M. and Cherdthong, A. 2009. Use of real-time PCR technique in studying rumen cellulolytic bacteria population as affected by level of roughage in swamp buffalo. Curr. Microbiol. 58: 294-299
- Zhou, M., Hernandez-Sanabria, E. and Guan, L.L. 2009. Assessment of the microbial ecology of ruminal methanogens in cattle with different feed efficiencies. Appl. Environ. Microbiol. 75: 6524-6533

Received on 07-01-2013 and Accepted on 11-03-2013